WOWII: Counterexamples to conjectures 63 and 85

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1 Conjectures

Conjecture 63 If G is a simple connected graph, then $f(G) \geq CEIL[(minimum\ of\ dist_{even}(v)+b(G)+1)/3]$

Conjecture 85 If G is a simple connected graph, then $tree(G) \geq CEIL[sqrt(1+2*minimum\ of\ dist_{even}(v))]$

Notations

- $f(G)$: The number of vertices of a largest induced forest of the graph.
- $CEIL[x]$: The smallest integer greater than or equal to x.
- minimum of $dist_{even}(v)$: even distance from a vertex v: The number of vertices whose distance from v is an even integer. The minimum of $dist_{even}(v)$ is the smallest among all $dist_{even}(v)$ for v a vertex of the graph. Note: the vertex v is itself counted, since $d(v, v) = 0$ is an even integer.
- $b(G)$: bipartite number of a graph.
- $tree(G)$: tree number of a graph.

2 Results

2.1 Graph identification

Both graphs invalidate Conjectures 63 and 85. They are defined by the following graph6 strings:

```
WO~~~~}?OH@aFAN@N_VwB~
```

```
Q~~~~~~_A?cBCFAF_bwC~?V{?~w
```

Property	Value
n	16
m	64
$f(G)$	4
$b(G)$	4
Minimum of $dist_{even}(v)$	8
$tree(G)$	4

Table 1: Properties of graph 1

Property	Value
n	18
m	81
$f(G)$	4
$b(G)$	4
Minimum of $dist_{even}(v)$	9
$tree(G)$	4

Table 2: Properties of graph 2

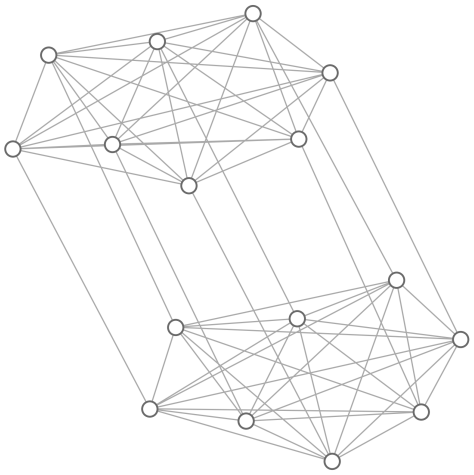


Figure 1: Graph 1

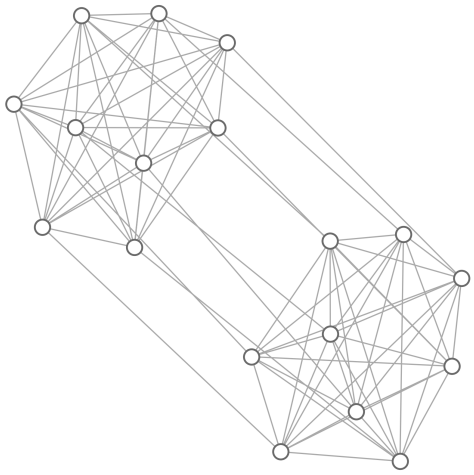


Figure 2: Graph 2

3 Verification of the inequality for conjecture 63

We detail the inequality for counterexample graph 1.

1. Calculation of the left-hand side (LHS):

$$\text{LHS} = f(G) = 4$$

2. Calculation of the right-hand side (RHS):

$$\text{RHS} = \left\lceil \frac{\min dist_{even}(v) + b(G) + 1}{3} \right\rceil$$

$$\text{RHS} = \left\lceil \frac{8 + 4 + 1}{3} \right\rceil = \lceil 4.333 \rceil = 5$$

Conclusion

The conjecture states that $f(G) \geq \text{RHS}$. However, we observe:

$$f(G) = 4 \quad \text{and} \quad \text{RHS} = 5$$

$$4 < 5$$

The inequality is not satisfied. **This graph is therefore a valid counterexample.**

4 Verification of the inequality for conjecture 85

We detail the inequality for counterexample graph 1.

1. Calculation of the left-hand side (LHS):

$$\text{LHS} = \text{tree}(G) = 4$$

2. Calculation of the right-hand side (RHS):

$$\text{RHS} = \left\lceil \sqrt{1 + 2 \times \min dist_{even}(v)} \right\rceil$$

$$\text{RHS} = \lceil \sqrt{1 + 2 \times 8} \rceil = \lceil \sqrt{17} \rceil = 5$$

Conclusion

The conjecture states that $\text{tree}(G) \geq \text{RHS}$. However, we observe:

$$\text{tree}(G) = 4 \quad \text{and} \quad \text{RHS} = 5$$

$$4 < 5$$

The inequality is not satisfied. **This graph is therefore a valid counterexample.**